

Introduction To LangSmith

Instructor

Dipanjan Sarkar

Head of Community & Principal AI Scientist at Analytics Vidhya

Google Developer Expert - ML & Cloud Champion Innovator

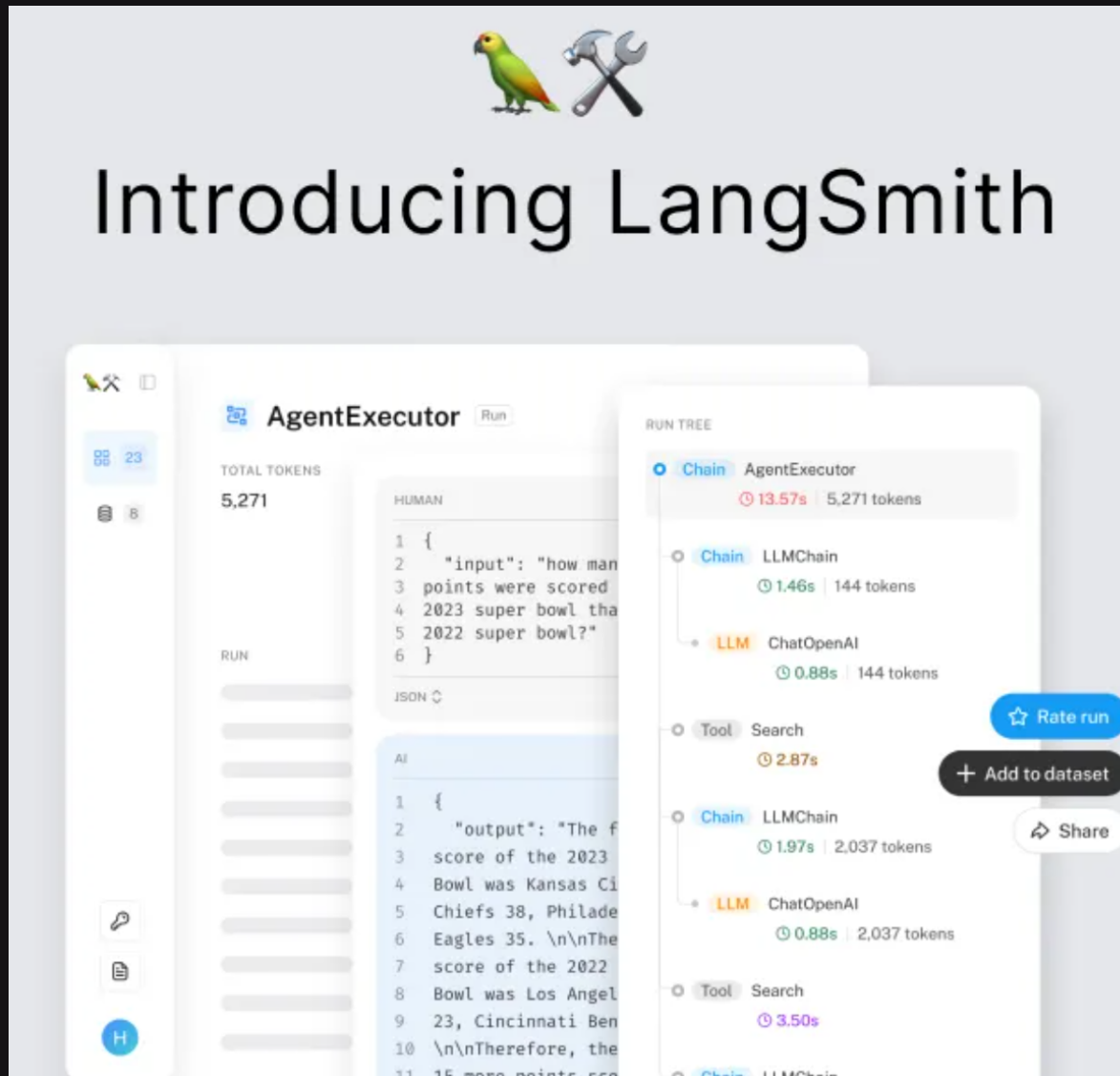
Published Author



Course Outline

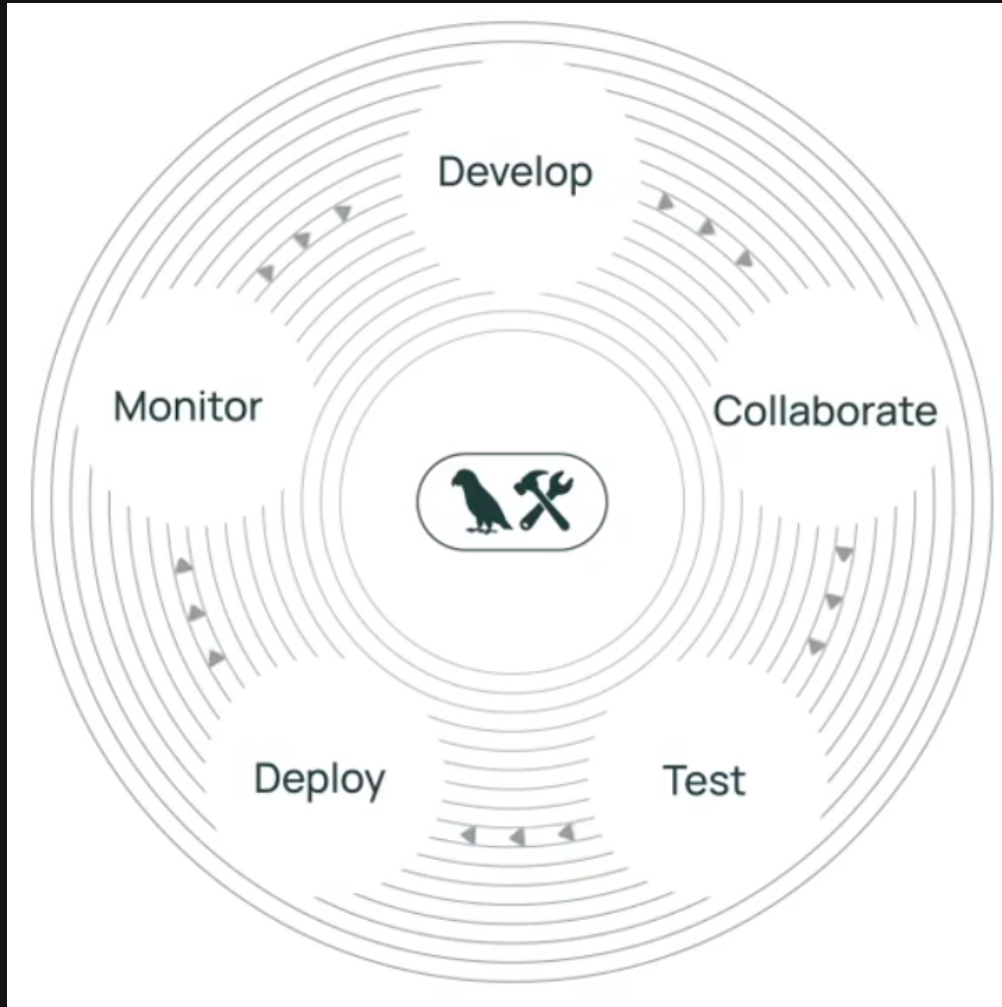
- Understanding LangSmith
- Develop with more visibility
- Collaboration with teammates
- Test and evaluate LLM apps
- Monitoring in production

Understanding LangSmith



- LangSmith is a unified platform from LangChain, for debugging, testing, evaluating, and monitoring LLM applications
- Help you easily track changes, debug, run, and evaluate any LLM application and agents

Understanding LangSmith



- LangSmith enables you to create a project, link into your LLM apps, and then develop, collaborate, test, deploy, and monitor these apps.

Develop With More Visibility

The screenshot displays the LangSmith interface for monitoring and debugging LLM applications. On the left, a 'TRACE' panel shows a sequence of steps in a chat application: 'chat-langchain' (5.13s, 5,846 tokens), 'FindDocs' (0.70s), 'RetrievalChainWith...' (0.68s), 'Retriever' (0.68s, highlighted), 'GenerateResponse' (5.07s), and 'ChatOpenAI' (4.33s). The 'Retriever' step is expanded on the right, showing its 'INPUT' and 'OUTPUT'.

Retriever
0.68s 5 DOCUMENTS 0

INPUT 09:01:00

How do I use a RecursiveUrlLoader to load content from a page?

OUTPUT 09:01:68

- Recursive URL | Langchain [Skip to main content](#__docu...
- langchain_community.document_loader.RecursiveUrlLoader - La...
- lazy_load() Lazy load web pages. load() Load web pages. load_An...
- python os.environment["LANGCHAIN_TRACING_V2" = "true" os....
- for result in results: print("Content:", result[0].page_content, "---...

At the bottom, there are three buttons: 'Add to Dataset', 'Annotate', and 'Playground'.

- LangSmith enables visibility into the entire sequence of calls
- Help find out errors and performance bottlenecks in the LLM applications
- Enable debugging and experimenting LLM apps with ease

Collaboration With Teammates

The screenshot displays the LangSmith 'ProductionTraces' interface. At the top, it shows summary statistics: 578 total runs, 23,010 total tokens, and latency metrics (P99 1.20s, P50 3.20s). Below this is a table of individual runs, each with a checkbox, name, input, latency, and feedback. A 'Filters' sidebar is open on the right, allowing users to filter runs by various criteria.

Run	Setup																				
☐	<table><thead><tr><th>Name</th><th>Input</th><th>Latency</th><th>Feedback</th></tr></thead><tbody><tr><td>☐ LLMChain</td><td>taylor swift annual tick...</td><td>3.20s</td><td>coherence 1.00</td></tr><tr><td>☐ LLMChain</td><td>superbowl points 2022</td><td>1.20s</td><td>conciseness 1.00</td></tr><tr><td>☐ LLMChain</td><td>favorite parrot phrase</td><td>2.50s</td><td>relevance 1.00</td></tr><tr><td>☐ LLMChain</td><td>best neighbourhood in SF..</td><td>7.60s</td><td>relevance 1.00</td></tr></tbody></table>	Name	Input	Latency	Feedback	☐ LLMChain	taylor swift annual tick...	3.20s	coherence 1.00	☐ LLMChain	superbowl points 2022	1.20s	conciseness 1.00	☐ LLMChain	favorite parrot phrase	2.50s	relevance 1.00	☐ LLMChain	best neighbourhood in SF..	7.60s	relevance 1.00
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Filters

Full-Text Search

Search...

Feedback

- ☒ Conciseness 24
- ☒ Correctness==1 24
- ☒ Relevance 1
- ☒ Harmfulness==0 24

Feedback Source

- ☐ App 24

Name

- ☐ LLM Chain 156

Run Type

- ☐ Chain 99

Status

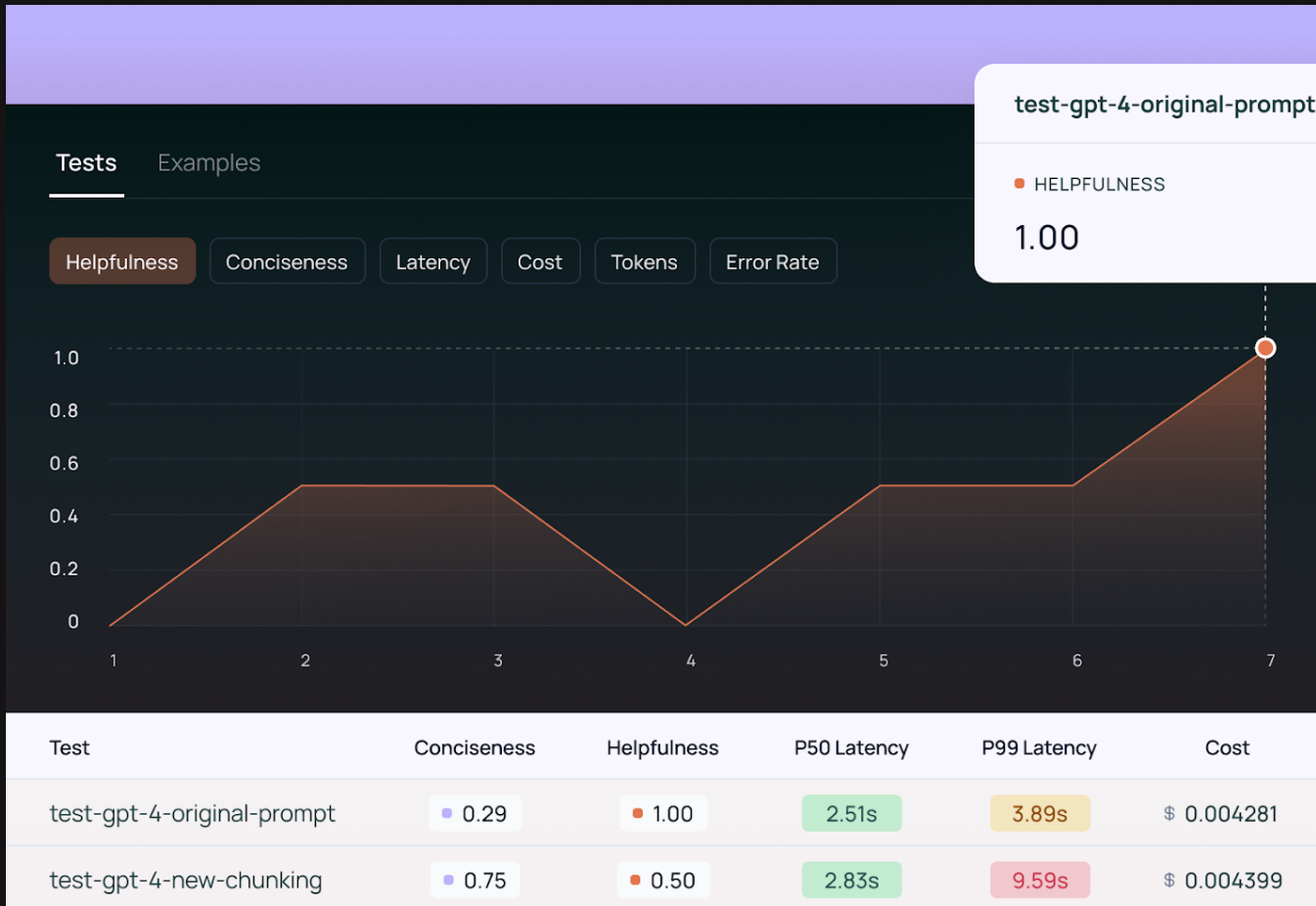
- ☐ Success 155
- ☐ Pending 1

Other

- ☐ Latency >= 10s
- ☐ Tokens >= 1,000

- LangSmith enables the sharing of useful insights about LLM applications within a team for collaborative development.
- Allows sharing of LLM app's traces (runs) within a team.
- Team can use the LangSmith Hub to craft, version, and collaborate on prompt templates.
- LangSmith also has Annotation Queues to add human labels and feedback on traces.
- LangSmith allows the construction of datasets from production data or existing sources for evaluations, few-shot prompting, fine-tuning, and more.

Test & Evaluate LLM Apps



- LangSmith enables human feedback on the LLM app or uses AI-assisted evaluation methods.
- Test the LLM app on reference LangSmith datasets.
- Supports both offline and online evaluation to test for problems and drift
- It contains many pre-built and custom evaluators to check relevance, correctness, harmfulness and more.

One-Click Deployment

chat-langchain

+ New Revision

Playground

Revision	Created	Actions
 870c2782-422a-44bc-923e-84712c918aba Building...	 2 min ago on  master	 Interrupt
 9cc5c887-9440-42d5-ba58-216793e2fccf Currently deployed	 1d ago on  master	 Redeploy
 f114e2c6-0936-40ab-a6b7-39f5aed37a18 Previously Deployed	 4d ago on  master	 Redeploy
 41cbe9fd-89fb-4e35-ada1-fad321686186 Revision interrupted	 5d ago on  master	 Redeploy
 f114e2c6-0936-40ab-a6b7-39f5aed37a18 Previously Deployed	 6d ago on  master	 Redeploy

- Easily deploy your LLM apps with hosted LangServe.
- Enables visibility into usage, errors, performance, and costs when you ship within the LangSmith platform.
- Built-in parallelization, fallbacks, batch, streaming, and async support.

Monitoring in Production



- LangSmith provides visibility on what's happening with the production application.
- Provides monitoring charts that allow tracking key metrics over time
- Allow to view charts such as Trace Latency, Tokens/Second, Cost, and feedback.

Thank You
